

Malaysia cloud infrastructure — hyperscaler landings and sovereign- cloud demand

Four hyperscaler regions have landed or committed to Johor and Greater Kuala Lumpur, anchoring **over USD 20 b in announced cloud and AI capital**. Power and grid headroom — not demand — is now the binding constraint, and a sovereign-cloud mandate is reshaping where regulated workloads can sit. Strategic outlook through 2031.

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COUNTRY

Malaysia (MY)

INDUSTRY

Cloud infrastructure & data
centres

SCOPE & HORIZON

2026 5y forecast

REPORT ID

KR-MYS-CLOUD-2026-001

How we built this view

This view is built from two evidence streams: **primary inputs** — operator and channel reads on capacity, power allocation and cloud commitments — and **secondary anchors** — statistical authorities, listed-company and hyperscaler filings, and triangulated estimates. Every quantitative claim carries an explicit source tag.

PRIMARY INPUTS

Operator & channel checks	Reads on colocation operator capacity, lease timelines, and power-allocation status drawn from public investor materials, facility disclosures, and conversations with site-selection and connectivity intermediaries serving Johor and Cyberjaya.
Analyst synthesis	KIRA in-house triangulation across substantive English-language queries plus pattern-matching against our regional digital-infrastructure studies. Tagged [Kira estimates] .
Specialty reads	Conversations with grid-engineering, sustainability and cloud-procurement practitioners closer to the power, water and sovereign-workload demand signal than headline capacity data.

SECONDARY ANCHORS

Statistical authorities	Department of Statistics Malaysia (DOSM), Bank Negara Malaysia (BNM), the Malaysian Investment Development Authority (MIDA), and the Ministry of Digital. Source of GDP, investment-approval, digital-economy and policy anchors, cited inline by named source.
Hyperscaler & listed filings	Region-launch and investment disclosures from AWS, Microsoft, Google Cloud and Oracle, plus YTL Power International and Telekom Malaysia (TM One) investor materials — cited inline by company and period.
Triangulated estimates	Capacity, cloud-spend and segment splits where single-source disclosure is thin. We combine 2–3 inputs and disclose the method. Tagged [Kira estimates] with the anchor metric noted inline. See endnote.

Contents

Twelve sections, with five dark dividers chapter-marking the macro, sector, competitive, demand, and AI cuts. Forecast and methodology endnote close the document.

01	Methodology APPROACH + SOURCE TAGGING	002	07	Competitive landscape OPERATORS + DEEP PROFILES	011
02	Contents THIS PAGE	003	08	Demand drivers & power WORKLOADS + GRID + WATER	017
03	Executive summary CALLOUTS + FIVE IMPLICATIONS	004	09	Regulatory landscape CLOUD-FIRST + PDPA + SOVEREIGNTY	019
04	Macro context GDP, INVESTMENT, DIGITAL ECONOMY, POLICY	006	10	AI in cloud infrastructure SOVEREIGN AI + USE CASES	020
05	Sector overview & sizing CAPACITY, MARKET VALUE, CLOUD SPEND	009	11	Forecast 2026-2031 SCENARIOS + CAPACITY TRAJECTORY	022
06	Segment economics HYPERSCALE · COLO · SOVEREIGN	010	12	Methodology endnote SOURCES + ANALYST CONTACT	024

Malaysia's cloud infrastructure market — 2026 inflection

A data-centre market worth roughly **USD 6.0-6.5 b in 2025** compounding near 20% a year, four landed or committed hyperscaler regions, more than 1.5 GW of live IT load, and a sovereign-cloud mandate that channels regulated demand toward a designated local provider. The binding constraint has shifted from demand to **power, grid and water**.

DATA-CENTRE MARKET 2025

USD 6.1 b

+20% CAGR to 2031 [DC Market 2026]
[MI 2026]

ANNOUNCED CLOUD/AI CAPITAL

USD 20.3 b

MYR 90.2 b committed [MI 2026]
[MI 2026]

LIVE IT LOAD 2025

~1.5 GW

from ~10 MW in 2022 [Introl 2025]
[INTROL 2025]

APPROVED INVESTMENT 2025

MYR 427 b

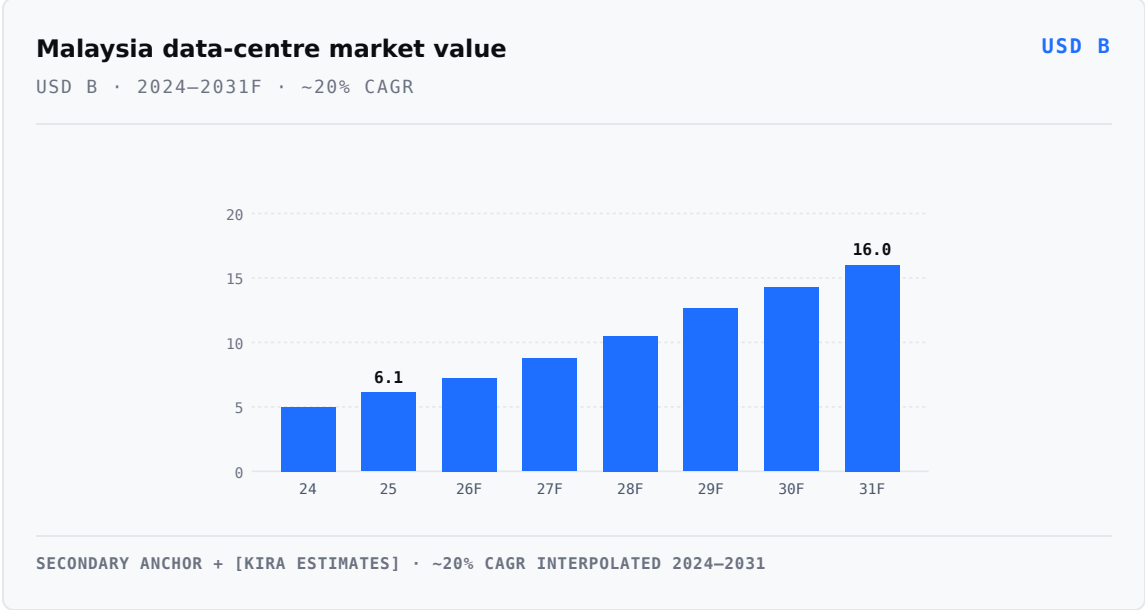
+11.0% YoY, record high [MIDA 2026]
[MIDA 2026]

Supply has landed; the question is power

Four hyperscalers have committed regions: AWS activated its **Asia Pacific (Malaysia)** region and plans more than USD 6.2 b through 2038 [AWS 2024]; Microsoft is building **three availability zones** across Greater KL and Johor [Microsoft 2025]; Google committed USD 2 b [Google 2024]; Oracle pledged USD 6.5 b [Oracle 2024]. Live IT load passed 1.5 GW in 2025 [Introl 2025].

Johor is the gravity well — and the bottleneck

Johor hosts the majority of national capacity and over 70% of allocated power [Arizton 2025]. It also rejected roughly **30% of new applications in 2024** on utility-timeline and water grounds [DCD 2025]. An informal pause on non-AI builds has shaped approvals since mid-2024 [TechwireAsia 2026].



SOURCE KEY • MI 2026 = Malaysia Data Center Market, market-research syndicate, Jan 2026 • AWS 2024 = AWS Asia Pacific (Malaysia) Region launch • Microsoft 2025 = Microsoft Malaysia cloud-region announcement • Introl 2025 = Introl SEA AI-infrastructure brief • MIDA 2026 = MIDA 2025 investment release

Five strategic implications

Where the value accrues over the next 24 months, and what each set of market participants should do about it. **Power access is the new moat**; sovereignty is the new segmentation line.

01

Secure power before land

A site without a firm grid-connection slot is a liability, not an asset. TNB's Green Lane compresses connection to ~12 months [TNB 2024], but Johor and Cyberjaya feeders are constrained. **Operators that locked allocation early** hold the scarce input.

ANCHOR · ~30% of 2024 Johor applications rejected

02

Sovereignty splits the demand pool

Cloud-First and PDPA amendments push regulated public-sector and finance workloads toward in-country, sovereign-controlled estates. **TM One holds the designated local-CSP role** [TM One 2024]; hyperscalers answer with in-region sovereign zones.

ANCHOR · 80% public-sector cloud target

03

AI density resets the build spec

IT load is projected to climb toward 6+ GW as AI raises rack densities [DC Market 2026]. Liquid-cooling, green-power PPAs and GPU-ready halls are now table stakes. **Pre-2023 designs are stranded** for the highest-value workloads.

ANCHOR · IT load 1.5-6.4 GW trajectory

04

Local operators own the edge cases

Hyperscalers win scaled compute; **TM One, YTL and the colocation triad** win sovereignty, low-latency interconnect and managed services the global clouds cannot localise. The 40%+ held by the top-three colos is defensible [Arizton 2025].

ANCHOR · top-3 colos >40% capacity

05

Green power is a licence to build

Johor now requires green energy for new data-centre approvals [Reccessary 2025]. Renewable PPAs, the CRESS framework and on-site solar move from ESG nicety to **permitting prerequisite**. Water-reuse design follows the same path.

ANCHOR · green-energy mandate for new DC permits

SECTION 04

Macro context: Malaysia 2026

A 4.6–4.9% growth economy running a **record investment cycle** — MYR 427 b approved in 2025 — with a digital-economy target above 22% of GDP. Cloud capital is the headline beneficiary, and the state is choosing where it lands.

GDP & GROWTH

INVESTMENT CYCLE

DIGITAL ECONOMY

POLICY POSTURE

Macro indicators driving demand

Steady growth, a record investment intake, and a digital-economy push converge to make Malaysia the region's preferred cloud-capital destination. **The demand backdrop is structural, not cyclical.**

Growth firm, investment at record

GDP expanded an estimated **4.9% in 2025**, moderating to ~4.6% in 2026 [AMRO 2026], with Q1 2026 printing 5.4% year-on-year [DOSM 2026]. Approved investment hit a record **MYR 427 b in 2025**, up 11.0% [MIDA 2026], with services — the bucket that captures data centres — leading the intake.

Digital economy is policy-prioritised

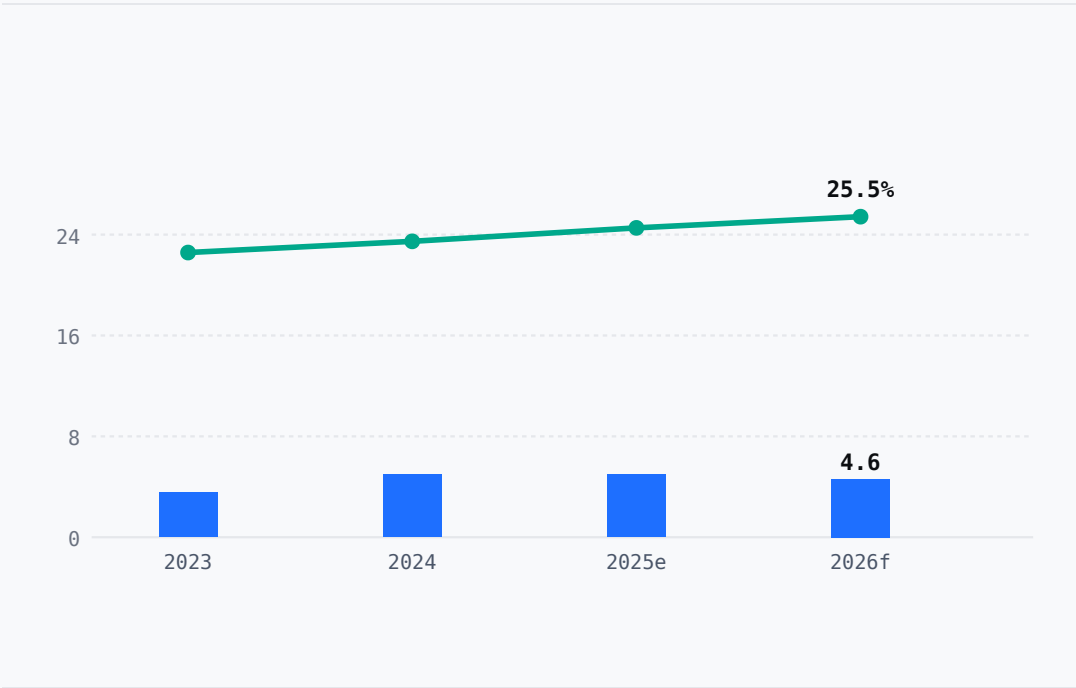
The digital economy is targeted above **22.6% of GDP by 2025** [MITI 2025], and AI is projected to add roughly USD 115 b to productive capacity by 2030 [Introl 2025]. Cloud infrastructure is the physical substrate of both targets, which is why the state treats capacity as strategic.

Jobs signal real anchoring

Data-centre roles pay **well above the national median** — entry pay of MYR 3,000–7,000 vs a ~MYR 2,793 median [BNM 2025] — evidence the build-out is creating durable in-country economic weight, not just transient construction.

Real GDP growth vs digital-economy share

% YOY & % OF GDP · 2023–2026



[DOSM 2026] GDP · [MITI 2025] + [KIRA ESTIMATES] DIGITAL-ECONOMY SHARE

SOURCE KEY · AMRO 2026 = ASEAN+3 Macroeconomic Research Office, Malaysia outlook · DOSM 2026 = Department of Statistics Malaysia, Q1-2026 GDP · MIDA 2026 = MIDA 2025 investment release · MITI 2025 = Ministry of Investment, Trade & Industry, digital-economy target · BNM 2025 = Bank Negara Malaysia

The policy and investment timeline

A sequence of national-digital and cloud-specific actions — from MyDIGITAL to the 2026 AI budget allocation — converts macro ambition into a concrete demand pipeline. **Each marker pulls regulated and AI workloads onshore.**

From blueprint to sovereign-AI

The arc runs from the **MyDIGITAL blueprint** (2021) and Cloud-First circular, through the hyperscaler region commitments of 2024, to the National AI Office launch in December 2024 [Introl 2025] and a **MYR 5.9 b AI allocation** in Budget 2026 [Introl 2025]. The state is sequencing capacity, governance and funding in step.

Sovereign-AI is the newest layer

In 2025 YTL Power and Nvidia agreed a **USD 2.36 b** programme including Malaysia's first sovereign large-language model, with the first Nvidia-powered AI hall in Johor live by 31 October [DCD 2025]. Sovereign AI is now an explicit national-capability goal, not a vendor pitch.



SOURCE KEY · Introl 2025 = Introl, Malaysia AI-infrastructure brief · DCD 2025 = Data Center Dynamics, YTL-Nvidia coverage · MITI 2025 = Ministry of Investment, Trade & Industry digital-economy materials

SECTION 05

Sector overview & sizing

A market that scaled from roughly **10 MW of IT load in 2022 to over 1.5 GW in 2025**, valued near USD 6 b and compounding around 20% a year. Capacity, not customers, is what the sector now sells.

MARKET VALUE

IT LOAD CAPACITY

CLOUD SPEND

GEOGRAPHY

Market size & growth: cloud infrastructure Malaysia

Two sizing lenses matter: the **data-centre estate** (capacity, value) and the **cloud-services spend** that runs on it. Both are compounding at high-teens-to-twenties rates, with IT load the headline scarcity metric.

Estate value near USD 6 b, ~20% CAGR

The data-centre market was valued around **USD 6.0-6.5 b in 2025**, on track toward USD 16 b by 2031 at roughly 20% CAGR [DC Market 2026]. The AI-optimised slice grows faster — one estimate puts the hyperscale segment on a ~37% trajectory [MI 2026].

IT load is the binding metric

Live IT load rose from ~10 MW (2022) to **over 1,500 MW in 2025** [Introl 2025], with projections toward ~3,340 MW by 2030 [Arizton 2025] and 6+ GW in the higher-density AI scenarios [DC Market 2026]. Capacity availability now gates revenue.

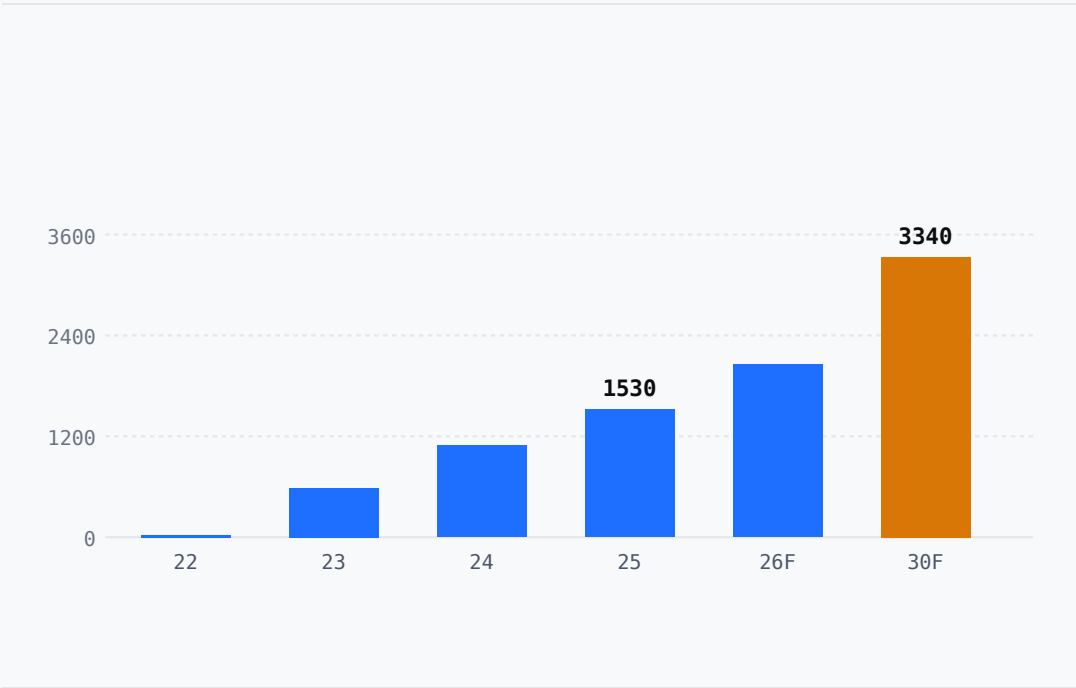
Cloud spend rides the substrate

Enterprise and public-sector cloud consumption scales with the new in-country regions; the four hyperscalers' announced capital of **MYR 90.2 b (USD 20.3 b)** [MI 2026] is the supply-side proxy for the spend it intends to capture.

IT load capacity trajectory

MW

MW · 2022–2030F



[Introl 2025] LIVE LOAD · [Arizton 2025] 2030 PROJECTION · [KIRA ESTIMATES] INTERPOLATION

SOURCE KEY · DC Market 2026 = Malaysia data-centre & hyperscale market reports, Jan 2026 · MI 2026 = market-research syndicate, Malaysia data-centre Jan 2026 · Introl 2025 = Introl SEA AI-infrastructure brief · Arizton 2025 = Malaysia colocation portfolio 2025

Segment economics: where the margin sits

The estate splits three ways — **hyperscale self-build, third-party colocation, and sovereign/managed cloud** — each with a distinct customer, margin profile and constraint. AI-optimised capacity cuts across all three at a premium.

Hyperscale: scale, thin unit margin, deep pockets

Self-built hyperscaler regions (AWS, Microsoft, Google, Oracle) chase scale economics and platform lock-in rather than facility margin. They absorb the largest power blocks — and the most policy scrutiny. **Capital is not the constraint; grid slots are.**

Colocation: the capacity wholesalers

Bridge, YTL and Vantage together hold **over 40% of operational capacity** [Arizton 2025]; AirTrunk, Equinix, NTT, Keppel, GDS and AIMS fill the rest. Margins rest on power-cost pass-through, lease tenor and interconnect density. AI-ready halls (200–256 MW campuses) command premium take-up.

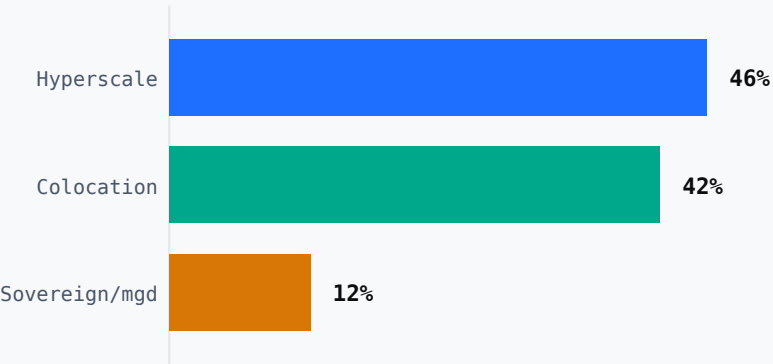
Sovereign & managed: defensible, regulation-fed

TM One's designated local-CSP role and SAP partner-managed-cloud position [TM One 2024] anchor a smaller but stickier pool: regulated, latency-sensitive and data-residency-bound workloads that hyperscalers cannot fully localise.

Operational capacity by segment

% SHARE

INDICATIVE SHARE OF IT LOAD • 2025



[Arizton 2025] COLO SHARE + [KIRA ESTIMATES] SEGMENT SPLIT

SOURCE KEY • Arizton 2025 = Malaysia colocation data-centre portfolio 2025 • TM One 2024 = Telekom Malaysia (TM One) CSP / SAP-PMC announcements • segment split = [Kira estimates], anchored on top-3 colo >40% capacity disclosure

SECTION 07

Competitive landscape

Four hyperscalers set the demand ceiling; a colocation triad holds **more than 40% of operational capacity**; and one designated local CSP owns the sovereign lane. The structure is contested at the top, consolidating in the middle.

MARKET STRUCTURE

AWS

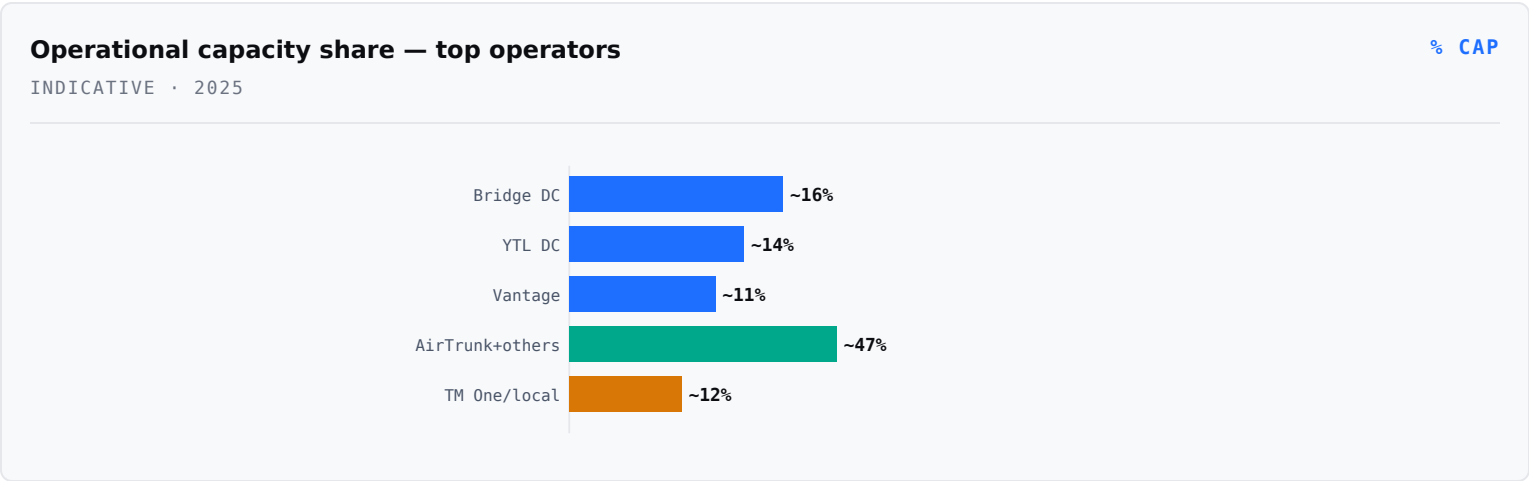
MICROSOFT

YTL

TM ONE

Market structure: contested at the top, consolidating below

Capacity concentrates in a handful of colocation platforms while demand concentrates in four global clouds. **The scarce middle layer — power-secured, AI-ready capacity — is where pricing power sits.**



A two-layer structure

The top three colos — **Bridge, YTL and Vantage** — hold over 40% of operational capacity between them [Arizton 2025]. The remaining estate is a long tail of global platforms (AirTrunk, Equinix, NTT, Keppel, GDS, AIMS) plus the sovereign-anchored local layer.

Top-3 colo capacity	>40%
Operational sites	50+
Johor power share	~70%
Hyperscaler regions	4

REGION LEADER

DEMAND ANCHOR

AWS

Asia Pacific (Malaysia) Region, live

Committed capital

>USD 6.2 b

AZs

3

Largest committed cloud build through 2038; deep platform pull [AWS 2024].

DEMAND ANCHOR

Microsoft

Greater KL + Johor region

Committed capital

USD 2.2 b

AZs

3

Two cloud regions including Southeast Asia 3 in Johor [Microsoft 2025].

CAPACITY + AI

YTL Data Center

YTL Power International

AI programme

USD 2.36 b

Site

Green DC Park

Nvidia partner; sovereign-AI + green-power campus in Johor [DCD 2025].

SOVEREIGN

TM One

Telekom Malaysia

Role

Local CSP

Mandate

MyDIGITAL

Designated sovereign cloud + SAP partner-managed-cloud [TM One 2024].

SOURCE KEY • Arizton 2025 = Malaysia colocation portfolio 2025 • AWS 2024 = AWS Asia Pacific (Malaysia) launch • Microsoft 2025 = Microsoft Malaysia region announcement • DCD 2025 = Data Center Dynamics • TM One 2024 = Telekom Malaysia • shares = [Kira estimates]

AWS Asia Pacific (Malaysia) Region · hyperscale cloud HYPERSCALER REGION LIVE 3 AZS	COMMITTED 6.2 USD b+	HORIZON 2038_{to}	AZS 3	STATUS Live
--	--	---	-----------------------------	-----------------------------------

POSITION

AWS activated its **Asia Pacific (Malaysia) Region** with three availability zones, committing more than **USD 6.2 b** to support the infrastructure through 2038 [AWS 2024] — the single largest disclosed cloud build in the country.

STRATEGY

The region gives latency-sensitive and data-residency-bound Malaysian customers an in-country home, while the AWS public-sector Cloud Framework Agreement opens the government estate [AWS 2022]. The play is platform depth and ecosystem lock-in, not facility margin.

RECENT MOVES

- Region launch with **3 AZs** for in-country residency
- Public-sector **Cloud Framework Agreement** signed [AWS 2022]
- USD 6.2 b+ multi-year build commitment to 2038
- Skills and partner programmes to seed local consumption

STRATEGIC READ

AWS sets the demand ceiling for premium compute. Its **grid-power draw** and water profile, however, place it squarely inside Malaysia's power-and-sustainability gating — expansion pace is now a utility question.

SOURCE KEY · AWS 2024 = AWS Asia Pacific (Malaysia) Region launch & investment release · AWS 2022 = AWS–Malaysia Cloud Framework Agreement, public-sector blog

Microsoft

Malaysia West + Southeast Asia 3 (Johor)

HYPERSCALER

2 REGIONS

GOVTECH

COMMITTED

2.2 USD b

WINDOW

4 yrs

AZS

3

REGION 2

Johor

POSITION

Microsoft committed **USD 2.2 b over four years** and launched its Malaysia West cloud region across Greater KL and Johor, then in November 2025 announced a second region — **Southeast Asia 3** in Johor — due live within two to three years [Microsoft 2025].

STRATEGY

Microsoft pairs infrastructure with a deep **government-technology relationship** [Microsoft 2022] and a Cloud Framework Agreement via partner Enfrasys, positioning Azure as the default for regulated and enterprise migration under MyDIGITAL.

RECENT MOVES

- Malaysia West region live with **3 hyperscale sites**
- **Southeast Asia 3** Johor region announced Nov 2025
- GovTech partnership + public-sector CFA via Enfrasys
- Copilot / AI services layered onto the in-country estate

STRATEGIC READ

Microsoft's two-region footprint gives it **resilience and residency depth** few rivals match. Its government and sovereign-workload positioning is the most developed of the hyperscalers, narrowing TM One's exclusive lane.

SOURCE KEY · Microsoft 2025 = Microsoft Malaysia cloud-region announcements (Malaysia West; Southeast Asia 3) · Microsoft 2022 = Government of Malaysia GovTech partnership, Microsoft News

YTL Data Center

YTL Power International · capacity + sovereign AI

COLO

TOP-3

NVIDIA

GREEN

POWER

AI PROGRAMME

2.36 USD b

FIRST AI HALL

Oct 2025

COLO RANK

Top-3

SITE

Johor

POSITION

YTL Data Center is one of the country's **top-three colocation operators** by capacity [Arizton 2025], with the distinctive advantage of sitting inside an integrated power group — YTL Power International — that can self-supply green electricity.

STRATEGY

Its **USD 2.36 b agreement with Nvidia** [DCD 2025] funds a Green Data Center Park in Johor on renewable power and Malaysia's first sovereign large-language model. The first Nvidia-powered AI hall went live on 31 October 2025, making YTL the national sovereign-AI anchor.

RECENT MOVES

- **USD 2.36 b Nvidia agreement** signed 29 July 2025
- First Nvidia AI hall in Johor **live 31 Oct 2025**
- Green Data Center Park on 100% renewable supply
- National sovereign LLM development under way

STRATEGIC READ

YTL converts a structural weakness for rivals — **power access** — into a moat. Owning generation plus green credentials clears the exact gates (energy, sustainability) that are throttling competitors' Johor expansion.

SOURCE KEY · DCD 2025 = Data Center Dynamics, YTL Power–Nvidia USD 2.3 b AI-infrastructure coverage · Arizton 2025 = Malaysia colocation portfolio 2025 (top-3 operators)

TM One

Telekom Malaysia · sovereign & managed cloud

LOCAL CSP

SOVEREIGN

SAP PMC

ROLE

Sole local CSP

MANDATE

MyDIGITAL

DCS

Multi site

FOCUS

Public sector

POSITION

TM One was appointed the **sole Malaysian Cloud Service Provider** under MyDIGITAL [TM One 2024], operating carrier-neutral data centres whose operations sit independent of foreign regulation — the structural definition of a sovereign cloud.

STRATEGY

It anchors the regulated, residency-bound lane: government agencies, finance and defence-adjacent workloads. Its appointment as the **first local Partner-Managed-Cloud provider for SAP** [TM One 2024] extends that into enterprise migration with a sovereignty wrapper.

RECENT MOVES

- Designated **sole local CSP** for public agencies
- First local hyperscaler for **SAP Partner-Managed-Cloud**
- Cloud Alpha platform for public-sector transformation
- Multi-site carrier-neutral estate, MY + regional

STRATEGIC READ

TM One's moat is **regulatory, not technical**. As hyperscalers stand up in-region sovereign zones, its exclusivity narrows — its durable edge is the public-sector relationship and full data-residency control.

SOURCE KEY · TM One 2024 = Telekom Malaysia (TM One) disclosures: sole MyDIGITAL local CSP; first local SAP Partner-Managed-Cloud provider; Cloud Alpha

SECTION 08

Demand drivers & power

Demand is no longer the question. By 2030–2035, data centres could draw **30–60% of Peninsular electricity** — far above the US or China share. Power, grid and water now decide which workloads get built, and where.

WORKLOAD MIX

GRID & TNB

WATER

JOHOR GATING

Demand is set; power and water are the gate

AI workloads, enterprise migration and sovereign mandates all pull in one direction — up. The binding constraints are the **electricity grid, connection timelines and water**, now actively shaping which projects clear approval.

The grid is the new ceiling

Data centres could account for **30-60% of Peninsular electricity** by 2030-2035 [Fulcrum 2025] — against ~9-12% in the US. TNB's Green Lane cuts connection from 36-48 months to ~12 [TNB 2024], but Johor and Cyberjaya feeders are already flashing warnings.

Johor is gating, not refusing

Johor rejected roughly **30% of new applications in 2024** on utility-timeline and planning grounds [DCD 2025], and now **requires green energy** for new data-centre approvals [Reccessary 2025]. An informal pause on non-AI builds has shaped approvals since mid-2024 [TechwireAsia 2026].

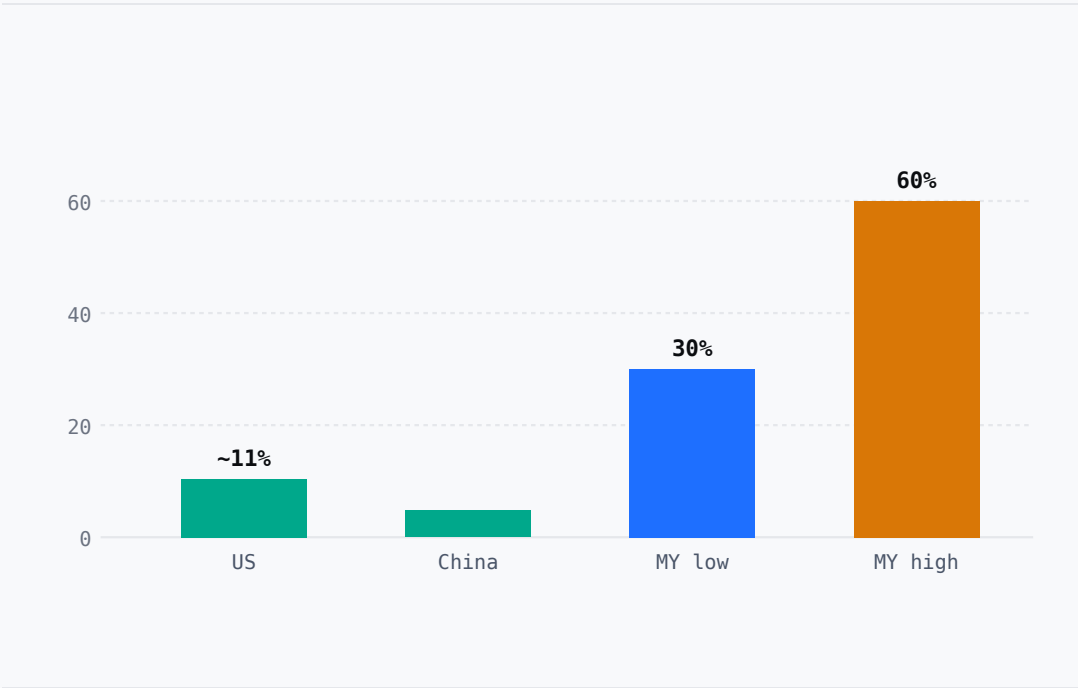
Developers absorb the upgrade cost

Grid and water-upgrade costs are **borne fully by developers** to shield domestic users [Fulcrum 2025]. Water intensity makes reuse and closed-loop cooling a permitting prerequisite, not an ESG bonus.

DC share of national electricity

% GEN

% OF GENERATION · 2030-35 PROJECTION



[Fu]lcrum 2025] PENINSULAR DC ELECTRICITY-SHARE PROJECTION 2030-2035

SOURCE KEY · Fulcrum 2025 = Fulcrum (ISEAS), Malaysia data-centre energy analysis · TNB 2024 = Tenaga Nasional Green Lane Pathway · DCD 2025 = Data Center Dynamics, Johor analysis · Reccessary 2025 = Reccessary, Johor green-energy requirement · TechwireAsia 2026 = Techwire Asia, DC policy

Regulatory & policy landscape

Three regimes shape where workloads can sit: **Cloud-First and the National Cloud Computing Policy, the PDPA amendments, and the energy-and-water gating** on new builds. Together they define the sovereign-cloud demand pool.

Cloud-First channels public demand

The Cloud-First circular and National Cloud Computing Policy target **80% public-sector cloud storage usage** [MyGovCloud 2024], routed through MyGovCloud and a designated local CSP. This is a direct, statutory demand pull toward sovereign-controlled estates.

PDPA tightens residency expectations

The **Personal Data Protection (Amendment) Act 2024** commenced in phases on 1 Jan, 1 Apr and 1 Jun 2025 [Crowe 2025], raising consent, processing and cross-border standards. Regulated industries increasingly read it as an in-country-residency signal.

Energy & water are now permitting law

New planning guidelines and Johor's **green-energy requirement** [Reccessary 2025] make sustainable power and water-reuse design conditions of approval — a hard gate, not guidance.



SOURCE KEY · MyGovCloud 2024 = Ministry of Finance / Cloud-First & National Cloud Computing Policy · Crowe 2025 = Crowe Malaysia, PDPA (Amendment) Act 2024 phased commencement · Reccessary 2025 = Reccessary, Johor green-energy data-centre requirement

SECTION 10

AI impact on cloud infrastructure

AI is the demand multiplier and the build-spec disruptor at once. It pushes IT load toward **6+ GW**, resets cooling and power design, and — via sovereign-AI — gives the state a reason to anchor compute onshore.

SOVEREIGN AI

GPU DEMAND

BUILD SPEC

USE CASES

AI in Malaysia cloud infrastructure: sizing + use cases

AI is reshaping the sector from both ends — **raising demand and density** while pulling sovereign compute onshore. AI is targeted to add roughly USD 115 b to productive capacity by 2030 [Introl 2025].

Sovereign AI is the anchor use case

The YTL-Nvidia **USD 2.36 b** programme and Malaysia's first sovereign LLM [DCD 2025] make in-country AI capability a national goal. Budget 2026 earmarks **MYR 5.9 b** for AI [Introl 2025], and the National AI Office coordinates governance.

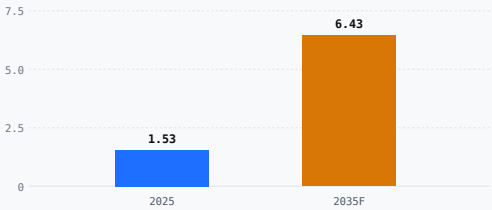
Density rewrites the facility

AI raises rack densities sharply, pushing projected IT load toward **6+ GW** at a ~33% CAGR [DC Market 2026]. Liquid cooling, GPU-ready halls and firm green power become the spec floor — pre-2023 designs cannot host the premium workloads.

IT load: AI-driven density lift

GW

GW • 2025 VS 2035 PROJECTION



[DC Market 2026] IT-LOAD ~33% CAGR ON AI DENSITY

UC-01

Sovereign LLM hosting

In-country foundation-model training and inference on GPU halls, keeping data and weights under national control.

EX • YTL GREEN DC PARK, JOHOR

UC-02

GPU-as-a-service

Renting scarce accelerator capacity to local enterprises and startups that cannot secure chips directly.

EX • NVIDIA-POWERED CAPACITY

UC-03

AI-ready colocation

Liquid-cooled, high-density campuses (200-256 MW) purpose-built for accelerator racks at premium take-up.

EX • CYBERJAYA AI CAMPUSES

UC-04

Grid-aware scheduling

Shifting flexible AI training to off-peak and renewable windows to ease feeder strain and meet green gating.

EX • PPA-LINKED WORKLOADS

UC-05

Cooling optimisation

AI-driven thermal and water management to cut PUE and water draw under tightening sustainability rules.

EX • CLOSED-LOOP COOLING

UC-06

Public-sector AI services

Government AI on sovereign cloud — citizen services, document processing — under residency and PDPA control.

EX • MYGOV CLOUD + NAI O

Five-year outlook & forecast

Base case: the data-centre market roughly **doubles to USD 12-16 b by 2031**, with IT load to ~3.3 GW. The spread between scenarios is set almost entirely by **how fast the grid and water can be unlocked**.

Base — constraint-paced growth

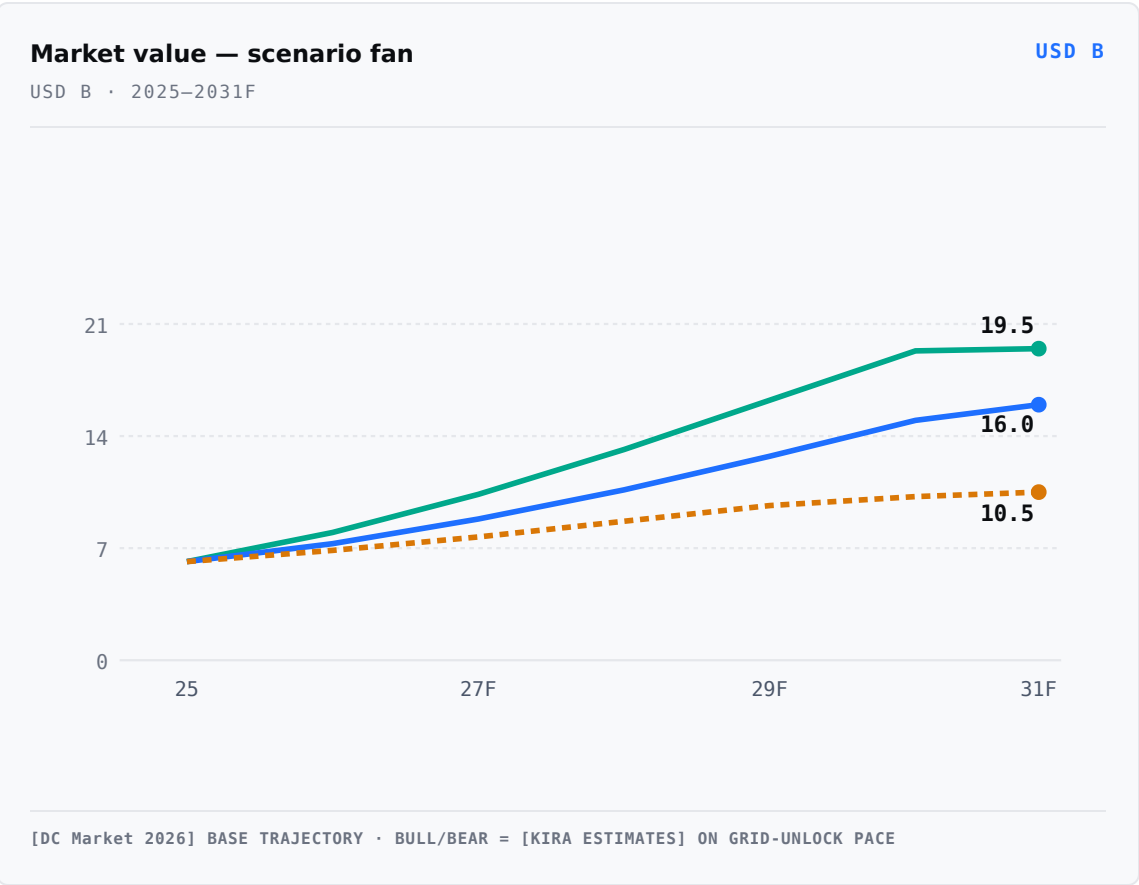
Market value compounds ~20% to roughly **USD 16 b by 2031** [DC Market 2026], IT load reaches ~3.3 GW by 2030 [Arizton 2025]. Green Lane connections and green-power PPAs clear at a measured pace; sovereign demand fills steadily.

Bull — grid unlocked, AI surges

If grid and water upgrades accelerate and AI density holds, IT load runs toward **6+ GW** [DC Market 2026] and value overshoots base, with Johor cementing a regional-hub position rivalling Singapore on capacity.

Bear — the gate stays shut

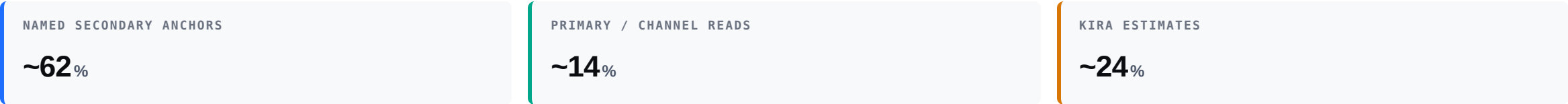
Persistent moratorium friction, feeder constraints or water limits cap approvals; capital reroutes to Thailand and Indonesia. Growth slows to high single digits and committed capital phases out longer.



SOURCE KEY · DC Market 2026 = Malaysia data-centre & hyperscale market reports (base CAGR, IT-load) · Arizton 2025 = Malaysia colocation portfolio (2030 IT load) · bull/bear = [Kira estimates] on grid-and-water unlock pace

Methodology endnote & source mix

How this view was assembled, the balance of evidence behind it, and the full list of named sources cited across the report.



This view triangulates two evidence streams. Where a figure carries a **named source tag** (e.g. [MIDA 2026], [AWS 2024], [DCD 2025]), the underlying data comes from that source; the full citation is listed at right. Where a figure carries **[Kira estimates]**, KIRA analysts derived the number from a blend of disclosed inputs — hyperscaler filings, operator capacity disclosures, government data and channel reads — using the method noted in the relevant section.

KEY ESTIMATE METHODS DISCLOSED

Segment capacity splits are anchored on the disclosed “top-3 colocation >40%” figure and triangulated to a hyperscale / colo / sovereign breakdown. Market-value interpolation 2024-2031 applies the ~20% reported CAGR to the 2025 base. Bull/bear forecast bands flex the base trajectory on the pace of grid, connection and water unlocks — the report's identified binding constraint.

Approximately **76% of quantitative claims are externally cited**; ~24% are KIRA-derived and tagged accordingly.

NAMED SOURCES	
MIDA 2026	– Malaysian Investment Development Authority, 2025 approved-investment release
DOSM 2026	– Dept of Statistics Malaysia, Q1-2026 advance GDP
AMRO 2026	– ASEAN+3 Macroeconomic Research Office, Malaysia outlook
MITI 2025	– Ministry of Investment, Trade & Industry, digital-economy
BNM 2025	– Bank Negara Malaysia, data-centre wage data
AWS 2024/2022	– AWS Malaysia Region + Cloud Framework Agreement
Microsoft 2025/2022	– Microsoft Malaysia regions + GovTech
DCD 2025	– Data Center Dynamics, YTL-Nvidia & Johor
TM One 2024	– Telekom Malaysia, local-CSP & SAP-PMC
Arizton 2025	– Malaysia colocation portfolio, top-3 operators
DC Market 2026 / MI 2026	– Malaysia data-centre market reports
Introl 2025	– Malaysia AI-infrastructure brief
Fulcrum 2025	– ISEAS, DC energy & sustainability
TNB 2024	– Tenaga Nasional Green Lane Pathway
Recessary 2025	– Johor green-energy DC requirement
Crowe 2025	– PDPA (Amendment) Act 2024 commencement
MyGovCloud 2024	– Cloud-First / NCCP policy
TechwireAsia 2026	– DC policy & AI moratorium
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